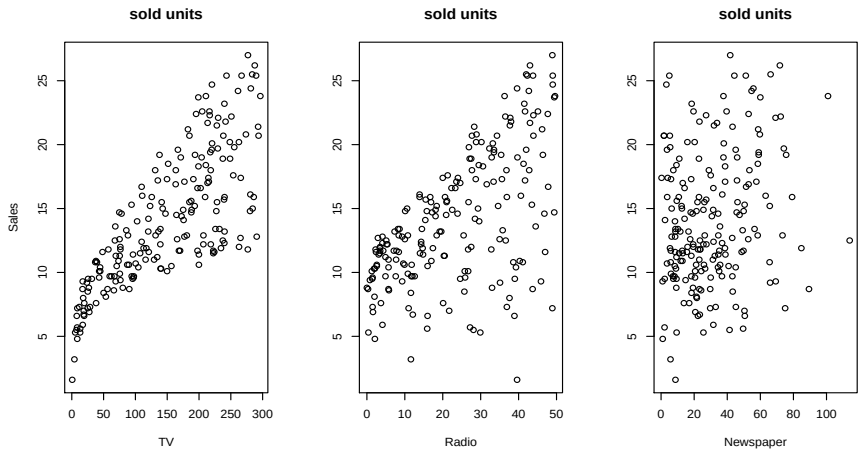


STA314, Lecture 1

September 14, 2021
Stanislav Volgushev

Motivating example (regression)

You work for a marketing firm. One of your clients collected data in 200 market locations. For each market, they have the amount of money spent on TV, Radio and newspaper advertisement (in 1000 \$) of a certain product and the sales (in thousands of units) of that product during a given time frame. They want you to help them predict sales and help decide how to advertise.



Some terminology

Typically, we will have a *data set (sample)* consisting of *data (observations)*
 $(x_1, Y_1), \dots, (x_n, Y_n)$

- ▶ x_i can be real values, vectors or other types of variables (images, videos, text...).
- ▶ In *regression* Y_i usually can take many different values, often on a continuous scale
- ▶ In *classification* (later in lectures) Y_i will take on a small number of different values. Sometimes those values are not numeric.

x_i are called *predictors (covariates, regressors, features, input variables)*.

Y_i are called *response (outcome, target, dependent variable)*.

What are the aims in Statistical Learning?

Prediction: Given a new predictor value x_0 , predict response Y_0 .

- ▶ Example: assume we spend 151 thousand \$ on TV advertisement for a certain market. How many units do we expect to sell?
- ▶ Example: assume we spend 120 thousand \$ on TV advertisement, 15 thousand \$ on radio advertisement and 5 thousand \$ on newspaper advertisement. How many units do we expect to sell?

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Inference: Understand the relationship between predictor and response in more detail and use that to make decisions. Quantify uncertainty.

- ▶ Example: does spending money of TV advertisement have any effect on sales?
- ▶ Example: assume we increase TV advertisement by 10 thousand \$. What is the effect on sales? How precisely can we quantify this effect?
- ▶ Example: assume we can spend a given budget on advertising on TV, radio and newspaper. How should we allocate the budget to maximise sales?

Regression models

Given observations $(x_1, Y_1), \dots, (x_n, Y_n)$ try to find relationship between x_i and Y_i
(example above: x_i is money spent on TV, Y_i are sales)

Machine Learning/Statistics: build a *regression model* to describe influence of x_i on Y_i .

Such models take the form (for the experts: this is the fixed design formulation)

$$Y_i = f(x_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0$$

where f is called *regression function* and ε_i are called *errors*.

- ▶ x_i are assumed to be fixed numbers.
- ▶ f describes the 'systematic' influence of x_i on Y_i .
- ▶ ε_i captures everything that f does not describe. ε_i is random variable.
- ▶ $\mathbb{E}[\varepsilon_i] = 0$ implies: $f(x_i)$ is the average value of Y_i at predictor value x_i .

$$\begin{aligned} &\Downarrow \\ &\mathbb{E}[Y_i] \\ &= f(x_i) \end{aligned}$$

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Remarks on ε_i

- ▶ In advertisement example: variation due to unobserved factors such as size of market, local population, 'randomness' (ex.: weather for umbrella sales) etc.
- ▶ In Physics experiments: measurement error
- ▶ In Biology: genetic variation, environmental influence etc.

How do we use this model for prediction?

$$Y_i = f(x_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0$$

A typical way of measuring the quality of a prediction $\hat{Y}_0$ at point x_0 is the expected *mean squared error* (short: **MSE**)

$$MSE(\hat{Y}_0) = \mathbb{E}[(Y_0 - \hat{Y}_0)^2].$$

If $Y_0 = f(x_0) + \varepsilon_0$, x_0 fixed number and ε_0 independent of $\hat{Y}_0$ then (see notes)

$$\mathbb{E}[(\hat{Y}_0 - Y_0)^2] = \mathbb{E}[(\hat{Y}_0 - f(x_0))^2] + \text{Var}(\varepsilon_0).$$

But before we derive this: which $\hat{Y}_0$ will minimize the MSE if f is known?

unknown

compute from data

$\hat{Y}_0 = f(x_0)$

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But before we derive this: which $\hat{Y}_0$ will minimize the MSE if f is known?

- ▶ $\text{Var}(\varepsilon_0)$ irreducible part. Even if we know $f(x_0)$ this part can not be improved.
- ▶ $\mathbb{E}[(\hat{Y}_0 - f(x_0))^2] \geq 0$ depends on $\hat{Y}_0$. $\mathbb{E}[(\hat{Y}_0 - f(x_0))^2] = 0$ if and only if $P(\hat{Y}_0 = f(x_0)) = 1$.
- ▶ MSE minimized at $\hat{Y}_0 = f(x_0) = \mathbb{E}[Y_0]$.
- ▶ If we know f our best prediction is $f(x_0)$. But we usually don't know f .

A first estimator for f : the K-nn method

The best possible value for $\hat{Y}_0$ is $\hat{Y}_0 = \mathbb{E}[Y_0] = f(x_0)$. Given data $(x_1, Y_1), \dots, (x_n, Y_n)$, how do we *learn/estimate* $f(x_0)$?

First: assume $x_1, \dots, x_n$ take only values 0 or 1 and $x_0 = 0$. *✓ Ave($Y_i : X_i = 0$)*

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$$\hat{f}(0) = \text{Average}(Y_i : x_i = 0) = \frac{\sum_{i=1}^n Y_i I\{x_i = 0\}}{\sum_{i=1}^n I\{x_i = 0\}}$$

Here

$$I\{x_i = 0\} = \begin{cases} 1, & \text{if } x_i = 0 \\ 0, & \text{otherwise} \end{cases}$$

$= \# \{x_i : x_i = 0\}$

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This works in data sets where $x_1, \dots, x_n$ take few distinct values and we are interested in predicting outcome for one of those values. Example: advertisement data set has no point with $x = 152$. Even if we had such a value this would be a bad prediction (why?).

TV

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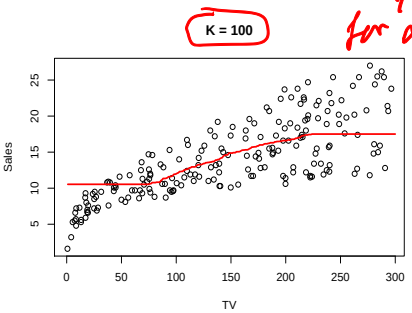
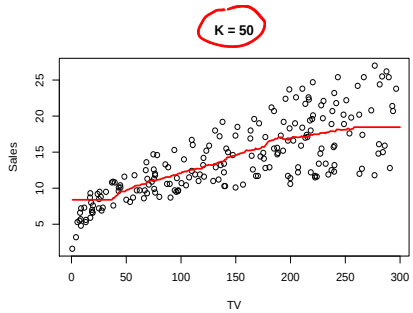
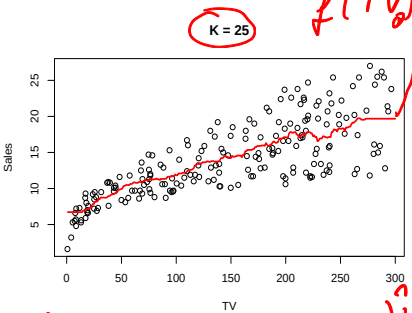
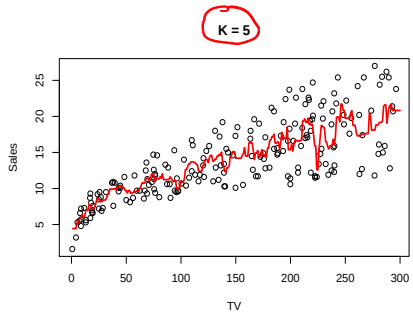
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Idea: instead of requiring $x_i = x_0$ take K of the 'closest' x_i . **K-nn** (K nearest neighbours) method.

$$\hat{f}(x_0) = \frac{\sum_{i=1}^n Y_i I\{x_i \text{ among closest } K \text{ to } x_0\}}{K}$$

Example: K-nn regression for predicting Sales from TV advertisement

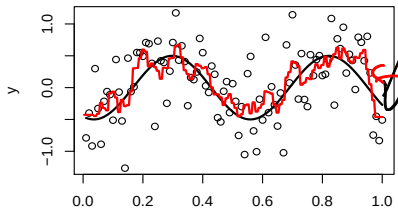


$f(TV)$

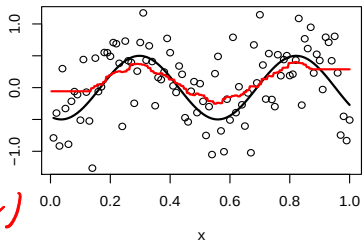
for diff. X

Examples of K-nn with simulated data

$f(x) = \sin(-2+12*x)$, $K = 5$



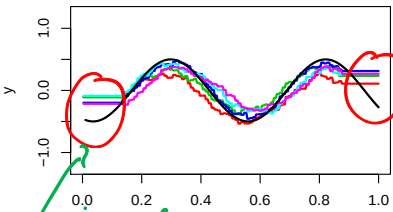
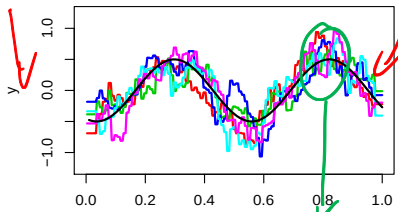
$f(x) = \sin(-2+12*x)$, $K = 25$



each curve: different ϵ_i
 $K=5$

$f(x)$
 $\downarrow$ K -nn

$K=25$



variance large bias large

Bias-variance decomposition

- ▶ For smaller K , regression function is very 'wiggly'. A lot of variation between samples generated from the same model.
- ▶ If K is very large, data are not described well. Less variation between samples generated from the same model.

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This can be formalized through the concepts of *bias* and *variance*. Recall that the ~~ir~~reducible part of the MSE takes the form $\mathbb{E}[(\hat{Y}_0 - f(x_0))^2]$. After some calculations

$$\mathbb{E}[(\hat{Y}_0 - f(x_0))^2] = \text{Var}(\hat{Y}_0) + \underbrace{\{\mathbb{E}[\hat{Y}_0] - f(x_0)\}^2}_{\text{bias}^2}$$

- ▶ $\mathbb{E}[\hat{Y}_0] - f(x_0)$ is called **bias**. It describes how far from the truth the prediction $\hat{Y}_0$ is *on average*.
- ▶ $\text{Var}(\hat{Y}_0)$ describes how much variation the estimator $\hat{Y}_0$ has.
- ▶ Ideal estimator would have small bias and small variance, but usually that is impossible.

The bias and variance of K-nn

$Y_i = f(x_i) + \varepsilon_i$, ε_i i.i.d. with $\mathbb{E}[\varepsilon_i] = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, x_i fixed, $\hat{f}(x_0)$ K-nn estimator. Then, K closest points x_i to x_0 are unique

$$\text{Var}[\hat{f}(x_0)] = \sigma^2/K,$$

Bias: more complicated. For simplicity: $x_i = i/n$, $i = 1, \dots, n$, $f : [0, 1] \rightarrow \mathbb{R}$ two times continuously differentiable, $K = 2\ell + 1$, $x_0 = j/n$ with $\ell < j < n - \ell$. Then

$$\mathbb{E}[\hat{f}(x_0)] = \frac{1}{K} \sum_{u=-\ell}^{\ell} f\left(\frac{j+u}{n}\right) \approx f(x_0) + \frac{1}{24} f''(x_0) (K/n)^2 + r_{K,n}$$

where $r_{K,n}$ remainder term, 'small' under suitable conditions (more details: homework).

- ▶ Variance decreases as K increases
- ▶ Approximate squared bias increases as K increases
- ▶ To get a good MSE we have to balance bias and variance. This is called *bias-variance trade-off*

*K ↗ Var ↓
Bias ↑*

Remark: if we let $\ell = \ell_n$ with $\ell_n/n \rightarrow 0$ in the analysis above this would be 'Asymptotic Statistics'. There will be some advanced homework related to that.